

Real Analysis II

MAT 2305

No of credits: 03

Pre requisite: MAT 1205

Contents:

Chapter 01 $\Rightarrow$ Infinite seriesChapter 02 $\Rightarrow$ Power SeriesChapter 03 $\Rightarrow$ Riemann IntegralChapter 04 $\Rightarrow$ Function of Several Variables (L)Chapter 05 $\Rightarrow$ Metric Spaces

Chapter 01 : Infinite Series

Grading

Tutorials: 20%

Mid Semester: 20%

End Semester: 60% (3h) Questions (6 or 5)

 $(a_n) \leftarrow$ Sequence $(S_n) \leftarrow$ A new sequence from (a_n)

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_4 = a_1 + a_2 + a_3 + a_4$$

$$S_n = a_1 + a_2 + a_3 + a_4 + \dots + a_n = \sum_{k=1}^n a_k$$

 $(S_n) \leftarrow$ A infinite series

Let $a_n = 1/n$ $n \in \mathbb{N}$

$$S_1 = a_1 = 1/n = 1$$

$$S_2 = a_1 + a_2 = 1/2 + 1 = 3/2$$

$$S_3 = a_1 + a_2 + a_3 = 1 + 1/2 + 1/3 = 11/6$$

$$S_4 = a_1 + a_2 + a_3 + a_4 = 1/3 + 1/2 + 1 + 1/4 = \frac{25}{12}$$

$$(S_n) = (S_1, S_2, S_3, S_4, \dots)$$

$$= (1, 3/2, 11/6, 25/12, \dots)$$

← A series.

$$(S_1, S_2, S_3, S_4, \dots) = \sum_{n=1}^{\infty} a_n$$

* Let (a_n) be a sequence of real numbers. we associate another sequence (S_n) with (a_n) such that

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

this is $= \sum_{k=1}^n a_k$ for all $n \in \mathbb{N}$

S_n is called the n^{th} partial sum of

* The sequence (S_n) is called an infinite series (or just a series) and it is denoted by the expression

$$\sum_{n=1}^{\infty} a_n$$

$$(S_n) = (a_1, a_1 + a_2, a_1 + a_2 + a_3, \dots, \sum_{k=1}^n a_k, \dots)$$

$\uparrow$
series

* S_n is called The n^{th} partial sum of the infinite series (S_n) .

The infinite series $\sum_{n=1}^{\infty} a_n$ is said to be convergent or be divergent

$$|a_n - L| < \epsilon \quad (1/n) \quad 1/n \rightarrow 0 \text{ as } n \rightarrow \infty \text{ converges to } 0$$

$1 \rightarrow$ converges to 1

$$\begin{aligned} a_n &\rightarrow L_1 & (-1)^n &\Rightarrow (-1, 1, -1, 1, \dots) \text{ divergent} \\ a_n &\rightarrow L_2 \\ L &= L_2 \end{aligned}$$

* If the sequence (S_n) converges to S , then S is called the sum of the infinite series and we write

$$\sum_{n=1}^{\infty} a_n = S \leftarrow \text{unique value}$$

$$(S_n) = a_1, \sum_{k=1}^2 a_k, \sum_{k=1}^3 a_k, \sum_{k=1}^4 a_k, \dots, \sum_{k=1}^{n+1} a_k, \sum_{k=1}^{\infty} a_k, \dots$$

infinites

$$\text{i.e. } \sum_{n=1}^{\infty} a_n = S \Leftrightarrow \lim_{n \rightarrow \infty} S_n \Rightarrow \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k$$

Example 01

prove that $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

$$S_1 = \frac{1}{2}$$

$$S_2 = \frac{1}{6}$$

let $a_n = \frac{1}{n(n+1)}$ for all $n \in \mathbb{N}$

Then $S_n = a_1 + a_2 + a_3 + \dots + a_n$

$$= \sum_{k=1}^n a_k$$

$$= \sum_{k=1}^n \frac{1}{k(k+1)}$$

$a_n = \frac{1}{n} - \frac{1}{n+1}$ (having decomposed into partial fractions)

$$a_1 = \frac{1}{2} = 1 - \frac{1}{2}$$

$$a_2 = \frac{1}{6} = \frac{1}{2} - \frac{1}{3}$$

$$a_3 = \frac{1}{12} = \frac{1}{3} - \frac{1}{4}$$

$$a_{n-2} = \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

$$a_n = \frac{1}{n} - \frac{1}{n+1}$$

Adding all the n^{th} equation

gives
$$\sum_{k=1}^n a_n = 1 - \frac{1}{n+1} = S_n$$

$$\lim_{n \rightarrow \infty} S_n = 1$$

* Therefore the series $\sum_{n=1}^{\infty} a_n$ is convergent and has the sum equal to 1.

i.e.
$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

Example 02

Consider
$$\sum_{n=1}^{\infty} 1$$

Let $a_n = 1 \quad \forall n \in \mathbb{N}$

Then

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_n \\ &= 1 + 1 + 1 + \dots + 1 \\ &= n \end{aligned}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} n = \infty$$

Therefore S_n is divergent and so $\sum_{n=1}^{\infty} 1$ is divergent.

Example 03

Consider $\sum_{n=1}^{\infty} 0$

Let $a_n = 0 \quad \forall n \in \mathbb{N}$

$$\begin{aligned} \text{Then } S_n &= a_1 + a_2 + a_3 + \dots + a_n \\ &= 0 + 0 + 0 + \dots + 0 \\ &= 0 \end{aligned}$$

i.e. $S_n = 0$

$$\lim_{n \rightarrow \infty} 0 = \lim_{n \rightarrow \infty} S_n = 0$$

$\therefore (S_n)$ is convergent and S_0

$$\sum_{n=1}^{\infty} 0 = 0$$

Example 04

Consider $\sum_{n=1}^{\infty} \frac{1}{n}$

$\sum_{n=1}^{\infty} \frac{1}{n}$ is called harmonic series

Let $a_n = \frac{1}{n} \quad \forall n \in \mathbb{N}$

$$\begin{aligned} \text{Then } S_n &= a_1 + a_2 + a_3 + \dots + a_n \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad \forall n \in \mathbb{N} \end{aligned}$$

* Note that the sequence (S_n) is not Cauchy

Cauchy Sequence

A sequence (a_n) is Cauchy if for any $\epsilon > 0$
 $\exists$ an N such that

$$|a_n - a_m| < \epsilon$$

whenever $m, n > N$ Here both $m, n \in \mathbb{N}$

Proof That sequence not Cauchy (S_n)

Claim

Take any two (+)ve integers
 m and n with $n > m$

Now

$$|S_n - S_m| = \left| \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m}\right) \right|$$

$$= \left| \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n} \right|$$

$$n > m$$

$$= \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n} > \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

$$\frac{1}{n} < \frac{1}{m}$$

$$m+1 < n$$

$$m+2 < n$$

$$m+3 < n$$

$$= (n-m) \frac{1}{n}$$

$$= 1 - \frac{m}{n}$$

$$\frac{1}{n} < \frac{1}{m+1}$$

$$\frac{1}{n} < \frac{1}{m+2}$$

$$\frac{1}{n} < \frac{1}{m+3}$$

That is

$$\text{i.e. } |S_m - S_n| \geq 1 - \frac{m}{n} \quad \text{for all}$$

$m, n \in \mathbb{N}$ with
 $n > m$

No: _____

$(\frac{1}{n})$ convergent, $\sum_{n=1}^{\infty} \frac{1}{n}$ divergent

Date: _____

As $n > m$

Let $n = 2m$

$$\text{Then } |S_n - S_m| \geq 1 - \frac{m}{n} = 1 - \frac{m}{2m} = \frac{1}{2} > 0$$

$\therefore (S_n)$ is not a Cauchy sequence. Hence S_n is not convergent. $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent

Remark

A sequence of real numbers is convergent iff it is Cauchy sequence

$$\begin{aligned} \text{Also } |S_{n+1} - S_n| &= (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1}) \\ &\quad - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}) \\ &= \frac{1}{n+1} > 0 \quad \forall n \in \mathbb{N} \end{aligned}$$

$\therefore S_n$ is a monotonic increasing and it is not bounded above sequence

That is (S_n) is increasing and divergent

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Theorem 01

If $\sum_{n=1}^{\infty} a_n$ converges then, $\sum a_n$ convergent $\lim_{n \rightarrow \infty} a_n = 0$

$$\lim_{n \rightarrow \infty} a_n = 0$$

Note

This theorem says that $\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$

Is the converse of the theorem true?

If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is convergent?

Answer is no.

Counter example

Consider ~~the~~ $\sum_{n=1}^{\infty} \frac{1}{n}$

Let $a_n = \frac{1}{n}$ for all $n \in \mathbb{N}$

$$\therefore \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

But $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (previous) ←

Proof of Theorem 01

Let $S_n = a_1 + a_2 + a_3 + a_4 + \dots + a_{n-1} + a_n$

$\underbrace{\hspace{15em}}_{S_{n-1}}$

$$S_n = S_{n-1} + a_n$$

$$\Rightarrow S_n - S_{n-1} = a_n$$

That is

$$a_n = S_n - S_{n-1}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (S_n - S_{n-1}) \\ &= \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1} \\ &= S - S \\ &= 0 \end{aligned}$$

Since $\sum_{n=1}^{\infty} a_n$ is convergent, S_n is convergent.

$$\text{Let } \lim_{n \rightarrow \infty} S_n = S$$

$$\boxed{\lim_{n \rightarrow \infty} S_n = S}$$

Test for Divergence

$\lim_{n \rightarrow \infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ is
divergent

Example 05 :

Consider $\sum_{n=1}^{\infty} n \sin(1/n)$

Let $a_n = n \sin(1/n) \quad \forall n \in \mathbb{N}$

$$\therefore \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} n \sin(1/n)$$

$$= \lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} \quad x = 1/n$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$= 1 \neq 0$$

Therefore the series $\sum_{n=1}^{\infty} n \sin(1/n)$ is divergent.

$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)$ is divergent.

Example 06

Consider $\sum_{n=1}^{\infty} (-1)^n$

Let $a_n = (-1)^n \quad \forall n \in \mathbb{N}$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^n \quad \text{D.N.E}$$

~~Thus~~ Therefore $\lim_{n \rightarrow \infty} a_n \neq 0$ Therefore

$\therefore \sum_{n=1}^{\infty} (-1)^n$ is divergent

Alternative method

$$S_n = a_1 + a_2 + a_3 + a_4 + \dots + a_n \quad \left[\begin{array}{l} 0 \text{ if } n \text{ is even} \\ -1 \text{ if } n \text{ is odd} \end{array} \right]$$

~~2024.01.30~~

2024.01.30

Example no 07:

— 9

Consider the geometric series with
Common ratio r

$$\sum_{n=1}^{\infty} r^{n-1}$$

$$a, ar, ar^2, ar^3, \dots$$

↑
First term

Common ratio

$$1 + r + r^2 + \dots$$

$$\text{Let } a_k = r^{k-1}, \text{ where } k = 1, 2, 3, \dots$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

$$= 1 + r + r^2 + \dots + r^{n-1}$$

$$a + ar + ar^2 + \dots + ar^{n-1}$$

$$\text{Summation of } n \text{ terms} = \frac{a(1-r^n)}{1-r}$$

$r \neq 1$

$$1 + r + r^2 + \dots + r^{n-1} = n \times 1$$

Date: ___/___/___

$$= 1 + r + r^2 + \dots + r^{n-1}$$

$$= \begin{cases} \frac{1-r^n}{1-r} & \text{if } r \neq 1 \\ n & \text{if } r = 1 \end{cases}$$

$|r| < 1$ Case 01

$$-1 < r < 1$$

$$\bullet \quad \underline{|r| < 1} \quad \sim -1 < r < 1 \sim r \in (-1, 1)$$

$$|r| = \begin{cases} r & \text{if } r \geq 0 \\ -r & \text{if } r < 0 \end{cases}$$



$$\boxed{\begin{array}{l} |r| < 1 \\ \lim_{n \rightarrow \infty} r^n = 0 \end{array}}$$

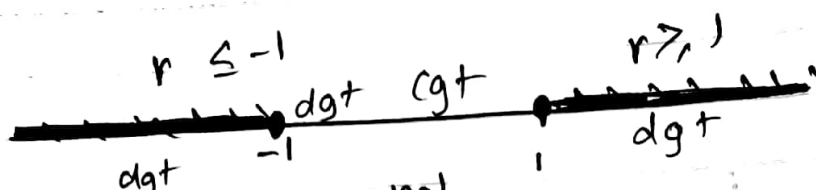
$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1-r^n}{1-r}$$

$$= \frac{1}{1-r}$$

$\therefore$ The geometric series is convergent.

$$\text{i.e. } \sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r} \quad \text{if } |r| < 1$$

Case 2 : $|r| \geq 1$



let $a_n = r^{n-1}$

$\lim_{n \rightarrow \infty} r^{n-1} \neq 0$

~~A_n~~ a_n convergent $\lim_{n \rightarrow \infty} a_n = 0$

$\lim_{n \rightarrow \infty} a_n = 0$ may or may not be convergent

Therefore $\sum_{n=1}^{\infty} a_n$ is not convergent
This is divergent

Example 08

Consider

$x = 0.999 \dots$

$x = 0 \times 10^0 + 9 \times 10^{-1} + 9 \times 10^{-2} + 9 \times 10^{-3} + \dots$

$= \frac{9}{10} \left(1 + \frac{1}{10} + \frac{1}{10^2} + \dots \right)$

geometric $r = \frac{1}{10}$

* This is geometric series with $r = \frac{1}{10}$

$$\frac{r}{1-r}$$

$$a \frac{1-r}{1-r}$$

$$a \frac{1-r}{1-r}$$

=

$$\frac{a/10}{1 - 1/10}$$

$$a/10 \left(\frac{1/10}{1 - 1/10} \right)$$

$$= \frac{a}{10} \times \frac{1}{10} (1-0) \cdot a \left(\frac{1-r^n}{1-r} \right)$$

$$= 1$$

$$= \frac{1}{10}$$

$$1 \times \frac{(1-0)}{(1 - \frac{1}{10})}$$

$$\frac{a}{10} \cdot \frac{1}{\frac{9}{10}}$$

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r} \quad \text{if } r \neq 1$$

$$a + ar + ar^2 + ar^3 + \dots = \lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r}$$

$$= \frac{a}{1-r} \lim_{n \rightarrow \infty} (1-r^n)$$

$$= \frac{a}{1-r} = \frac{a}{1-r}$$

$$x = 0.999 \dots$$

$$x = 0 \times 10^0 + 9 \times 10^{-1} + 9 \times 10^{-2} + 9 \times 10^{-3} + \dots$$

$$= \frac{9}{10} \left(1 + \frac{1}{10} + \frac{1}{10^2} + \dots \right)$$

$$= \frac{9}{10} \left(1 - \frac{1}{10} \right)$$

$$= \frac{9}{10} \times \frac{1}{9/10}$$

ഇതുകൊണ്ട്
അതിന്റെ
മൂല്യം

$$= \frac{a}{1-r}$$

This series converges to 1

$$\lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r}$$

Theorem 02

a) If the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converges then $\sum_{n=1}^{\infty} (a_n + b_n)$ converges and $a_n + b_n$ equals to $\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$.

b) If the series $\sum_{n=1}^{\infty} a_n$ converges

* and c is a constant, then the series $\sum_{n=1}^{\infty} (c a_n)$ is converges and

$$\sum_{n=1}^{\infty} (c \cdot a_n) = c \sum_{n=1}^{\infty} a_n$$

Proof.

Let s_n , t_n and u_n be the partial sums of $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} (a_n + b_n)$

$$s_n = a_1 + a_2 + a_3 + a_4 + \dots + a_n$$

$$t_n = b_1 + b_2 + b_3 + b_4 + \dots + b_n$$

$$u_n = c_1 + c_2 + c_3 + c_4 + \dots + c_n$$

when $c_k = a_k + b_k$

$$k=1, 2, 3, \dots$$

$$U_n = (a_1 + b_1) + (a_2 + b_2) + \dots + (a_n + b_n)$$

$$= (a_1 + a_2 + a_3 + \dots + a_n) + (b_1 + b_2 + b_3 + \dots + b_n)$$

$$U_n = S_n + t_n$$

We know that both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

$\uparrow$ $\uparrow$
 (S_n) (t_n)

Therefore the sequences S_n and t_n are convergent

$\therefore$ The sequence $(S_n + t_n)$ is also convergent

$$\text{Therefore } \lim_{n \rightarrow \infty} (S_n + t_n) = \lim_{n \rightarrow \infty} S_n + \lim_{n \rightarrow \infty} t_n$$

$$\text{where } = a + b \text{ and}$$

$$a = \lim_{n \rightarrow \infty} S_n, \quad b = \lim_{n \rightarrow \infty} t_n$$

$$\therefore c = \lim_{n \rightarrow \infty} U_n = a + b$$

$\therefore \lim_{n \rightarrow \infty} (a_n + b_n)$ is convergent

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

b)



Example 09

Find $\sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}}$

Note.

$$\frac{2^n - 4^{n+2}}{5^{n-1}} = 2 \left(\frac{2}{5} \right)^{n-1} - 4^3 \left(\frac{4}{5} \right)^{n-1}$$

Both the series $\sum_{n=1}^{\infty} \left(\frac{2}{5} \right)^{n-1}$ and $\sum_{n=1}^{\infty} \left(\frac{4}{5} \right)^{n-1}$

are geometric series with common

ratron. $r = \frac{2}{5}$ and $r = \frac{4}{5}$ Since $\frac{2}{5} < 1$ and $\frac{4}{5} < 1$

their common ratios are ~~less~~ in $(-1, 1)$

They are convergent

$\frac{a}{1-r}$

$$\sum_{n=1}^{\infty} \left(\frac{2}{5} \right)^{n-1} = \frac{1}{1 - \frac{2}{5}} = \frac{5}{3}$$

$$\sum_{n=1}^{\infty} \left(\frac{4}{5} \right)^{n-1} = \frac{1}{1 - \frac{4}{5}} = 5$$

$$\begin{aligned} \therefore \sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}} &= 2 \sum_{n=1}^{\infty} \left(\frac{2}{5} \right)^{n-1} - 4^3 \sum_{n=1}^{\infty} \left(\frac{4}{5} \right)^{n-1} \\ &= 2 \times \frac{5}{3} - 64 \times 5 \\ &= \frac{10}{3} - 320 \\ &= \frac{10 - 960}{3} = \frac{-950}{3} \end{aligned}$$

$\epsilon > 0$
 $\exists n \in \mathbb{N}$

x_n is bounded if there exists $M > 0$
s.t. $|x_n| < M$

Theorem 03

Let $\sum_{n=1}^{\infty} a_n$ be an infinite series of non negative terms. ($a_n \geq 0$)

Then, $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence of partial sums is bounded.

i.e. Let $S_n = \sum_{k=1}^n a_k$ iff $\equiv$ if and only if
 $\sum_{n=1}^{\infty} a_n$ is convergent iff $\equiv \iff$

$\{S_n\}$ is bounded.

ex: $1/n$ is bounded. $1/n \leq 1$

Proof

$$\text{Let } S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k \quad \forall n \in \mathbb{N}$$

$$\textcircled{1} \text{ gives } S_{n+1} = a_1 + a_2 + a_3 + \dots + a_n + a_{n+1}$$

$$S_{n+1} = S_n + a_{n+1}$$

$$S_{n+1} - S_n = a_{n+1} \geq 0 \quad \forall n \in \mathbb{N}$$

Therefore

$$S_{n+1} - S_n \geq 0$$

That is $S_{n+1} \geq S_n \quad \forall n \in \mathbb{N}$

$$S_1 \leq S_2 \leq S_3 \leq S_4 \leq \dots$$

So, the sequence (S_n) is monotonically increasing

* By the Monotonic Convergent theorem

(MCT) S_n is convergent iff (S_n) is bounded.

i.e. $\sum_{n=1}^{\infty} a_n$ is converge ~~iff~~ if

and only if bounded (bdd)
(at least bounded)

Theorem 04

* The infinite series $\sum_{n=1}^{\infty} a_n$ converges

iff and only if for each $\epsilon > 0$

$\exists$ a positive integer $N(\epsilon)$ s.t

$$|a_{n+1} + a_{n+2} + \dots + a_m| < \epsilon$$

where $m > n > N(\epsilon)$

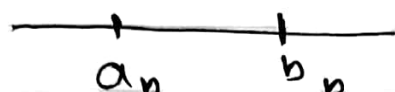
Theorem 05 (Comparison test)

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite

series of non-negative terms a_n and b_n with $a_n \leq b_n$

- a) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges
 b) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

$$a_n, b_n \geq 0 \quad \text{and} \quad a_n \leq b_n$$



$\sum a_n$ convergent $\Rightarrow$ $\sum b_n$ convergent

$\sum a_n$ divergent $\Rightarrow$ $\sum b_n$ divergent

$$a_n \leq \frac{1}{n} \quad b_n \geq 1$$

$\sum_{n=1}^{\infty} \frac{1}{n}$ divergent $\sum_{n=1}^{\infty} 1$ divergent

$$\frac{1}{n} \leq 1$$

$$a_n \leq \frac{1}{n^2} \quad \text{and} \quad b_n \geq 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} 1$$

Convergent

divergent

$$a_n \leq b_n$$

Example 09

Show that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent

Let $a_n = \frac{1}{n^2} \quad \forall n \in \mathbb{N}$

$$a_n = \frac{1}{n^2} = \left(\frac{1}{n}\right)\left(\frac{1}{n}\right) < \frac{1}{n} \left(\frac{1}{n-1}\right)$$

Let $b_n = \frac{1}{n(n-1)}$ for $n \geq 2$

$$\frac{1}{n(n-1)} = \frac{A}{n} + \frac{B}{n-1}$$

$$1 = A(n-1) + Bn$$

~~0 = 1~~

$$1 = B$$

$$1 = -2A$$

$$A = -\frac{1}{2}$$

$$1 = -A$$

$$A = -1$$

$$b_n = \frac{1}{n(n-1)} = \frac{-1}{n} + \frac{1}{n-1}$$

$$b_2 = \frac{-1}{2} + \frac{1}{2-1} = \frac{-1}{2} + 1 = \frac{1}{2}$$

$$b_3 = \frac{-1}{3} + \frac{1}{3-1} = \frac{-1}{3} + \frac{1}{2} = \frac{-2+3}{6} = \frac{1}{6}$$

$$b_2 = \cancel{-\frac{1}{2}} + \frac{1}{1}$$

$$b_3 = \cancel{-\frac{1}{3}} + \cancel{\frac{1}{2}}$$

$$\vdots$$

$$b_{n-1} = \cancel{-\frac{1}{n-1}} + \cancel{\frac{1}{n}}$$

$$b_n = \cancel{-\frac{1}{n}} + \frac{1}{n+1}$$

$$\sum_{k=2}^n b_k = 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{k=2}^n b_k = \lim_{n \rightarrow \infty} 1 - \frac{1}{n}$$

$$\Rightarrow \sum_{n=2}^{\infty} b_n = 1$$

$$\therefore \sum_{n=1}^{\infty} b_n = b_1 + \sum_{n=2}^{\infty} b_n = b_1 + 1 = 2$$

$\therefore$ Therefore $\sum_{n=1}^{\infty} b_n$ is converges,

$\therefore$ By OT $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is cgt



06.02.2023

$$a_n \leq b_n$$

$$c_n^+ \quad c_n^+$$

Example 10

Show that $\sum_{n=1}^{\infty} \frac{2^n}{3^n(n+1)^2+9}$ is convergent

Let

$$a_n = \frac{2^n}{3^n(n+1)^2+9} \geq 0 \text{ for all } n \in \mathbb{N}$$

$$a_n = \frac{2^n}{3^n(n+1)^2+9} < \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n \left(\frac{2}{3}\right)^{n-1}$$

$$= \frac{2}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1}$$

A geometric series with
common ratio $r = \frac{2}{3} < 1$

$\therefore$ It is convergent

Now by then, $\sum_{n=1}^{\infty} a_n$ is convergent

Example 11

Consider $\sum_{n=1}^{\infty} \frac{1}{5n-3}$

$$\text{Let } a_n = \frac{1}{5n-3} > 0 \quad \forall n \in \mathbb{N}$$

$$\frac{1}{5n-3} \leq \frac{1}{5n} \leq \frac{1}{7} > \frac{1}{10} \quad a_n \text{ cgt} \leq (b_n) \text{ cgt}$$

Note that $\frac{1}{5n-3} > \frac{1}{5n} \quad \forall n \in \mathbb{N}$

Since

$$\sum_{n=1}^{\infty} \frac{1}{5n} \quad \text{and} \quad b_n = \frac{1}{5n-3} \quad \forall n \in \mathbb{N}$$

Since $\sum_{n=1}^{\infty} \frac{1}{5n}$ divergent, $\sum_{n=1}^{\infty} a_n$ is divergent.

$\therefore \sum_{n=1}^{\infty} b_n$ is divergent.

Example 12

Consider $\sum_{n=1}^{\infty} \frac{1}{2n-1}$.

$$\text{Let } a_n = \frac{1}{2n-1} \geq 0 \quad \forall n \in \mathbb{N}.$$

Note that $\frac{1}{2n-1} > \frac{1}{2n} \quad \forall n \in \mathbb{N}$

$$\text{Let } a_n = \frac{1}{2n} \quad \text{and} \quad b_n = \frac{1}{2n-1} \quad \forall n \in \mathbb{N}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{2n}$ is divergent, $\sum_{n=1}^{\infty} a_n$ is divergent.

$\sum_{n=1}^{\infty} b_n$ is divergent.

* Example 12

Consider $\sum_{n=1}^{\infty} \frac{1}{2n-15}$

Let $a_n = \frac{1}{2n-15}$

$$\sum_{n=1}^{\infty} a_n = (a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 + \dots) + \sum_{n=8}^{\infty} \frac{1}{2n-15}$$

$$a_n = \frac{1}{2n-15}$$

$$a_1 = \frac{-1}{13} < 0$$

$$a_2 = \frac{-1}{11} < 0$$

$$a_3 = \frac{-1}{9} < 0$$

$$a_4 = \frac{-1}{7} < 0$$

$$a_5 = \frac{-1}{5} < 0$$

$$a_6 = \frac{-1}{3} < 0$$

$$a_7 = -1 < 0$$

$$a_8 = 1 > 0$$



No:

finite part + $\sum_{n=8}^{\infty} a_n$ Date:

Note that $\frac{1}{2n-15} > 0 \quad \forall n \geq 8$

$$\frac{1}{2n-15} > \frac{1}{2n} \quad \forall n \geq 8$$

$$= a_n'$$

Let $a_n' = \frac{1}{2n}$ Then $a_n > a_n'$

Since $\sum_{n=1}^{\infty} a_n'$ is divergent by CT,

$$\sum_{n=8}^{\infty} a_n \text{ is divergent}$$

Therefore, $\sum_{n=1}^{\infty} a_n$ is divergent

Definition

If $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent.

Example 13

Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

Let $a_n = \frac{(-1)^n}{n^2}$ for all $n \in \mathbb{N}$

$$\Rightarrow |a_n| = \left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2} \quad \left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2}$$

$$(-1)^n \leftarrow -1$$

We know that series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent

$\therefore$ The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent.

Theorem 10

If a series is absolutely convergent then it is convergent

$$\sum_{n=1}^{\infty} |a_n| < \infty \Rightarrow \sum_{n=1}^{\infty} a_n < \infty$$

$$\sum_{n=1}^{\infty} a_n < \infty \not\Rightarrow \sum_{n=1}^{\infty} |a_n| < \infty$$

Example 14

Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

We shall later show that the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \text{ is } < \infty$$

Let $a_n = \frac{(-1)^n}{n}$

Then $|a_n| = \frac{(-1)^n}{n} = \frac{1}{n}$

Also we know that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent i.e. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is not absolutely convergent.

but NOT ~~absolutely convergent~~ the series,

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is cgt. It is absolutely cgt

Theorem 7 (Limit form of CT).

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite series of positive terms

i.e. $a_n, b_n > 0$ such that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l$

If $0 < l < \infty$, then two series converge or diverge together

$$\begin{array}{lcl} \sum a_n & & \sum b_n \\ \text{cgt} & \Rightarrow & \text{cgt} \\ \text{cgt} & \Leftarrow & \text{cgt} \\ \text{dgt} & \Rightarrow & \text{dgt} \end{array}$$

$$\text{dgt} \Leftarrow \text{dgt}$$

Example 15

Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3 + 9}$

Let $a_n = \frac{n}{8n^3 + 9} > 0 \quad \forall n \in \mathbb{N}$

Note

$$\frac{1}{n^2} > \frac{n}{8n^3} > \frac{n}{8n^3 + 9} = a_n$$

Let $b_n = \frac{1}{n^2} > 0 \quad \forall n \in \mathbb{N}$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \left[\frac{\frac{n}{8n^3+9}}{\frac{1}{n^2}} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{n^3}{8n^3+9}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{8 + \frac{9}{n^3}}$$

$$= \frac{1}{8} > 0$$

① Also the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent

② Therefore by comparison test the series

$$\sum_{n=1}^{\infty} \frac{n}{8n^3+9} \text{ is } \text{cgt}$$

Example 1b

Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3-9}$

Let $a_n = \frac{n}{8n^3-9} > 0 \quad \forall n \geq 2$

$$\frac{1}{n^2} > \frac{n}{8n^3-9}$$

$$a_1 = \frac{1}{-1} = -1$$

$$a_n < b_n$$

$$b_n = \frac{1}{n^2} > 0 \quad \forall n \in \mathbb{N}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \left(\frac{n}{8n^3 - 9} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{n^3}{8n^3 - 9}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{8 - \frac{9}{n^3}}$$

$$\sum_{n=1}^{\infty} a_n = a_1 + \sum_{n=2}^{\infty} a_n > 0$$

$$\frac{n}{8n^3 - 9} > \frac{n}{8n^3} = \frac{1}{8n^2} \quad \text{--- (a.k.a) say}$$

$$\frac{1}{8n^2} \times \frac{8n^3 - 9}{n} \quad \parallel \quad \frac{1}{8n^2} \quad \parallel \quad a_n$$

$$b_n$$

$$L = \lim_{n \rightarrow \infty} \frac{1}{8n^2}$$

$$\frac{n}{8n^3 - 9}$$

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{8n^3 - 9}{8n^3} \\ &= \lim_{n \rightarrow \infty} \frac{8 - \frac{9}{n^3}}{8} \end{aligned}$$

$$a_n > 0, \quad n \geq 2 \quad (*)$$

But we know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is not $(*)$

$\therefore$ By $(*)$, the given series nonvergent.

Determine the following series are
Cgt or divg

(i) $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

(ii) $\sum_{n=1}^{\infty} \frac{\sin(n\theta)}{n^2}$ when $\theta \in \mathbb{R}$

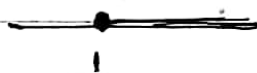
* Theorem 8 (Integral test)

Let f be a non negative ^{decreasing} continuous function define on $[1, \infty)$. Then the series $\sum_{n=1}^{\infty} f_n$ converges if and only if $\lim_{n \rightarrow \infty} \frac{1}{n} \int_1^n f(x) dx$ exist as

$$\lim_{n \rightarrow \infty} \int_1^n f(x) dx \text{ exist as}$$

$\in (-\infty, \infty)$
a real number

(i) : $f(x) \geq 0$ for all $x \in [1, \infty)$



(ii) f is ^(decreasing) on $[1, \infty)$

(iii) f is ~~contin~~ continuous $[1, \infty)$

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 2n}$$

$$f_n = \frac{1}{n^2 + 2n}$$

$$f(x) = \frac{1}{x^2 + 2x}$$

Example 17 $[1, \infty)$

Use the integral test to show that

$\sum_{n=1}^{\infty} 1/n$ is divergent

non negative.

$$\lim_{n \rightarrow \infty} \int_1^n 1/x \, dx$$

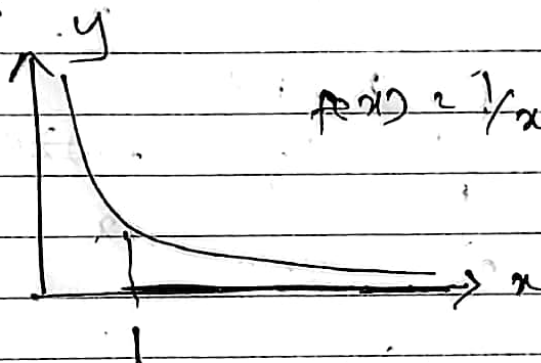
$$f(x) = \frac{1}{x} \quad x > 0$$

$1/x$
decreasing
continuous

$$\lim_{n \rightarrow \infty}$$

$$\left[\ln x \right]_1^n$$

$$f(x) = 1/x \, dx$$



The

(i) $f(x) > 0 \quad \forall x \in [1, \infty)$

(ii) $f(x)$ is decreasing on $[1, \infty)$

(iii) $f(x)$ is continuous $[1, \infty)$

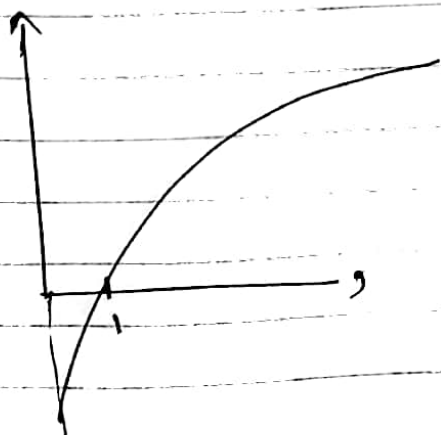
Therefore, we may apply the integral test

$$\lim_{n \rightarrow \infty} \int_1^n f(x) \, dx = \lim_{n \rightarrow \infty} \left[\int_1^n \frac{1}{x} \, dx \right]$$

$$= \left[\lim_{n \rightarrow \infty} \left[\ln|x| \right] \right]_{x=1}^{x=n}$$

$$= \lim_{n \rightarrow \infty} [\ln n - \ln 1]$$

$$= \infty$$



Thus, by the Integral test the given series is divergent.

Example 10

Consider the series $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$

$$\lim_{n \rightarrow \infty} \int_2^n f(x) dx$$

2 * $f(x) > 0$ $x \in [2, \infty)$ * ~~$x \in [2, \infty)$~~
 n * decreasing $[2, \infty)$

$$\lim_{n \rightarrow \infty} \int_2^n \frac{1}{x \ln x} dx, \quad x > 1$$

$$\lim_{n \rightarrow \infty} \int_2^n \frac{1/x}{\ln x} dx$$

$$\lim_{n \rightarrow \infty} \left[\ln(\ln x) \right]_2^n$$

$$\lim_{n \rightarrow \infty} [\ln(\ln n) - \ln(\ln 2)]$$

∞

By the Integral test the given series is divergent.

H.W

1) Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where

$p \in \mathbb{R}$

This is called a p series. Use the integral test to determine the convergent or divergent

(i) case 1: $p < 0$

(ii) case 2: $p = 0$

(iii) Case 3: ~~$p > 0$~~ $0 < p < 1$

(iv) Case 4: $p = 1$ (harmonic)

(v) Case 5: $p > 1$

2) Use p series test to determine the convergence of the following

(i) $\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

3) Show that the series

$\sum_{n=1}^{\infty} \frac{(-1)^n (n^2 + 3)}{(n^6 + 5n^3 + 7)}$

use p series
+ alt
limit form
is not

4) Use the integral test to determine the convergence / divergent

$\sum_{n=2}^{\infty} \frac{1}{n \ln(n) \ln(2n)}$

Theorem 09 (Ratio test)

Let $\sum a_n$ be a series with non zero terms

a) If $\exists$ a positive number r and positive integer $N \in \mathbb{N}$ such that

$$\left| \frac{a_{n+1}}{a_n} \right| \leq r < 1$$

for $n \geq N$,

then $\sum a_n$ is ~~absolutely~~ absolutely convergent

That means it is convergent

b) $\exists$ a number $r \geq 1$ and a positive ~~integer~~ integer $N \in \mathbb{N}$ s.t

$$\left| \frac{a_{n+1}}{a_n} \right| \geq r \geq 1 \text{ for } n > N,$$

then $\sum a_n$ is divergent

Theroem of (Limit form of ratio test)

Let $\sum_{n=1}^{\infty} a_n$ be a series with non zero terms. Let $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$.

- If $L < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (that is, convergent).
- If $L > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- If $L = 1$ then the test is inconclusive.

2024/02/13

Example 19

consider $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3^n}{n!}$

Let $a_n = \frac{(-1)^{n-1} 3^n}{n!}$ for all $n \in \mathbb{N}$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{(-1)^n 3^{n+1}}{(n+1)!} \cdot \frac{n!}{(-1)^{n-1} 3^n} \right| \\ &= \left| \frac{(-1)^n 3^{n+1}}{(n+1) \times n!} \times \frac{n!}{(-1)^{n-1} 3^n} \right| \\ &= \left| \frac{-3}{n+1} \right| = \frac{3}{n+1} \end{aligned}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3}{n+1} \right)$$

$$= 0 < 1$$

$\therefore$ from the limit form of the ratio test, the given series is absolutely convergent and hence it is convergent.

Example 20

Consider $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!}$

Let $a_n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} \quad \forall n \in \mathbb{N}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)(2n+1)}{(n+1)!} \times \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)} \right|$$

$$= \left| \frac{2n+1}{n+1} \right| = \frac{2 + \frac{1}{n}}{1 + \frac{1}{n}}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{2 + \frac{1}{n}}{1 + \frac{1}{n}} \right|$$

$$= 2 > 1$$

$\therefore$ The series is divergent according to the limit form of the ratio test.

Example 21

Consider $\sum_{n=1}^{\infty} (-1)^n$.

Let $a_n \in (-1)^n \quad \forall n \in \mathbb{N}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(-1)^{n+1}}{(-1)^n} \right|$$

$$= |-1|$$

$$= 1$$

$$\therefore L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\lim_{n \rightarrow \infty} 1$$

$\therefore$ LRRT cannot be applied.

But we know that $\sum_{n=1}^{\infty} (-1)^n$ is divergent

Example 22.

Consider $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Let $a_n = \frac{1}{n^2} \quad \forall n \in \mathbb{N}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} \right| = \left| \frac{n^2}{(n+1)^2} \right|$$

$$= \frac{n^2}{(n+1)^2} = \frac{n^2}{n^2 + 2n + 1} = \frac{1}{1 + \frac{2}{n} + \frac{1}{n^2}} \left[\frac{1}{1 + \frac{2}{n} + \frac{1}{n^2}} \right]^2.$$

$$\therefore L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{1 + 1/n} \right)^2$$

L.F.T can't be applied.

$$= 1$$

But we know that $\sum_{n=1}^{\infty} 1/n^2$ is convergent

Theorem 11 - Alternative Series test - AST

Let (C_n) be a decreasing sequence of positive numbers. Also suppose that the limit $C_n = 0$ as $n \rightarrow \infty$

Then the series,

$$\sum_{n=1}^{\infty} (-1)^{n-1} C_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^n C_n$$

converge.

$$* (C_n) \text{ is } \Rightarrow C_{n+1} - C_n \leq 0 \quad \forall n \in \mathbb{N}$$

$$C_{n+1} \leq C_n \quad C_n = 1/n \text{ d.}$$

$$* C_n > 0 \quad \forall n \in \mathbb{N}$$

$$* \lim_{n \rightarrow \infty} C_n = 0$$

Example 23

Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

Let $C_n = \frac{1}{n} > 0 \quad \forall n \in \mathbb{N}$

$$C_{n+1} - C_n = \frac{1}{n+1} - \frac{1}{n} = \frac{n - n - 1}{n(n+1)} = \frac{-1}{n(n+1)} < 0$$

$\therefore C_n$ is a decreasing sequence

$$\lim_{n \rightarrow \infty} C_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$\therefore$ We can apply the ~~test~~ AST.

Hence the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent

Root test

Defⁿ

Let (x_n) be a sequence of Real numbers

Then,

$\limsup x_n$ and $\liminf x_n$ are defined by

$$\limsup x_n = \inf \left(\sup_{k \geq n} x_k \right) = \inf \{ \sup \{ x_n, x_{n+1}, \dots \} \}$$

and

$$\liminf x_n = \sup \left(\inf_{k \geq n} x_k \right) = \sup \{ \inf \{ x_n, x_{n+1}, \dots \} \}$$

respectively

$$\text{Let } M_n = \sup \{x_n, x_{n+1}, x_{n+2}, \dots\}$$

$$M_1 = \sup \{x_1, x_2, x_3, \dots\}$$

$$M_2 = \sup \{x_2, x_3, x_4, \dots\}$$

$$M_3 = \sup \{x_3, x_4, x_5, \dots\}$$

$$\limsup x_n = \inf \{M_1, M_2, M_3, \dots\}$$

$$\text{Also let } m_n = \inf \{x_n, x_{n+1}, x_{n+2}, \dots\}$$

$$m_1 = \inf \{x_1, x_2, x_3, \dots\}$$

$$m_2 = \inf \{x_2, x_3, x_4, \dots\}$$

$$m_3 = \inf \{x_3, x_4, x_5, \dots\}$$

$$\liminf x_n = \sup \{m_1, m_2, m_3, \dots\}$$

for Example

$$\text{Let } x_n = \frac{1}{n}$$

$$M_n = \sup \{x_n, x_{n+1}, x_{n+2}, \dots\}$$

$$M_1 = \sup \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} = 1$$

$$M_2 = \sup \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} = \frac{1}{2}$$

$$M_3 = \sup \{\frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\} = \frac{1}{3}$$

$$\limsup x_n = \inf \{M_1, M_2, M_3, \dots\} = \inf \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$$

$$= 0$$

$$m_n = \inf \{ x_n, x_{n+1}, x_{n+2}, \dots \}$$

$$m_1 = \inf \{ 1, 1/2, 1/3, 1/4, \dots \} = 0$$

$$m_2 = \inf \{ 1/2, 1/3, 1/4, \dots \} = 0$$

$$m_3 = \inf \{ 1/3, 1/4, 1/5, \dots \} = 0$$

$$\liminf x_n = \sup \{ 0, 0, 0, 0, \dots \} = 0$$

Find $\limsup x_n$ and $\liminf x_n$ when $x_n = (-1)^n$ $\forall n \in \mathbb{N}$

$$M_n = \sup \{ x_n, x_{n+1}, x_{n+2}, \dots \}$$

$$M_1 = \sup \{ -1, 1, -1, 1, \dots \} = 1$$

$$M_2 = \sup \{ 1, -1, 1, -1, \dots \} = 1$$

$$M_3 = \sup \{ -1, 1, -1, \dots \} = 1$$

$$\limsup x_n = \inf \{ 1, 1, 1, 1, \dots \} = 1$$

$$m_n = \inf \{ x_n, x_{n+1}, \dots \}$$

$$m_1 = \inf \{ -1, 1, -1, 1, \dots \} = -1$$

$$m_2 = \inf \{ 1, -1, 1, -1, \dots \} = -1$$

$$\liminf x_n = \sup \{ -1, -1, -1, \dots \} = -1$$

Let x_n be a sequence of real numbers.
 Let $y_n = \sup \{x_n, x_{n+1}, x_{n+2}, \dots\}$

$$= \sup_{k \geq n} x_k$$

$$k \geq n.$$

Then

$$\limsup x_n = \lim_{n \rightarrow \infty} y_n \text{ and}$$

$$\text{ie } \limsup x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k$$

$$\text{Let } z_n = \inf x_k = \inf \{x_n, x_{n+1}, x_{n+2}, \dots\}$$

$$\liminf x_n = \lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} \left[\inf_{k \geq n} x_k \right]$$

$$x_n = 1/n$$

$$y_n = \sup \{1/n, 1/(n+1), 1/(n+2), \dots\} = 1/n$$

$$\limsup x_n = \lim_{n \rightarrow \infty} 1/n = 0$$

$$z_n = \inf \{1/n, 1/(n+1), 1/(n+2), \dots\} = 0$$

$$\liminf x_n = \lim_{n \rightarrow \infty} 0 = 0$$

Theorem 12 - Root test

Let $\sum a_n$ be a given series and let

$$\alpha = \limsup |a_n|^{1/n}$$

(i) If $\alpha < 1$ then $\sum a_n$ is absolutely convergent

(ii) If $\alpha > 1$ the $\sum a_n$ is divergent

(iii) If $\alpha = 1$ then the test is inconclusive
(cannot be applied)

Example 24

Consider $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n$

$$\text{Let } a_n = (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n$$

$$|a_n| = \left| (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^n \right| = \left| (-1)^{n+1} \right| \left| \left(\frac{2n+7}{8n-2} \right)^n \right|$$

$$\Rightarrow |a_n|^{1/n} = \frac{2n+7}{8n-2} \cdot \left(\frac{2n+7}{8n-2} \right)^{n/n}$$

$$\alpha = \limsup |a_n|^{1/n} = \limsup \left(\frac{2n+7}{8n-2} \right)$$

$$= \sup \left\{ \frac{2n+7}{8n-2} \mid n \in \mathbb{N} \right\} = \sup \left\{ \frac{9}{16}, \frac{11}{14}, \frac{13}{12}, \dots \right\}$$

$$Y_n = \sup \{ x_n, x_{n+1}, x_{n+2}, \dots \}$$

$$= \sup \left\{ \frac{2n+7}{8n-2}, \frac{2(n+1)+7}{8(n+1)-2}, \dots \right\}$$

$$\alpha = \limsup |a_n|^{1/n} = \limsup \left(\frac{2n+7}{8n-2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{2n+7}{8n-2} \right)$$

$$= \frac{1}{4}$$

$$\alpha = \frac{1}{4} < 1$$

$\therefore$ by the RT the series a_n is absolutely convergent

Example 25

Consider $\sum_{n=1}^{\infty} \left[\frac{3+(-1)^n}{3+2(-1)^n} \right]^n$

Let $a_n = \left[\frac{3+(-1)^n}{3+2(-1)^n} \right]^n \quad \forall n \in \mathbb{N}$

$$|a_n| = \left| \frac{3+(-1)^n}{3+2(-1)^n} \right|^n$$

$$|a_n|^{1/n} = \frac{3+(-1)^n}{3+2(-1)^n} \begin{cases} \frac{4}{5} & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd} \end{cases}$$

$$\alpha = \limsup |a_n|^{1/n} = \limsup \left(\frac{3+(-1)^n}{3+2(-1)^n} \right)$$

$$2 > 1$$

By the RT, the series is divergent

Example 26

Consider $\sum_{n=1}^{\infty} \left[\frac{2 + (-1)^n}{3 + 2(-1)^n} \right]^n$

Let $a_n = \left[\frac{2 + (-1)^n}{3 + 2(-1)^n} \right]^n$

$|a_n| = \left[\frac{2 + (-1)^n}{3 + 2(-1)^n} \right]^n$

$|a_n|^{1/n} = \frac{2 + (-1)^n}{3 + 2(-1)^n} \begin{cases} 3/5 & \text{if } n \text{ is even} \\ 1/3 = 1 & \text{if } n \text{ is odd} \end{cases}$

$\alpha = \limsup |a_n|^{1/n} = \limsup \left(\frac{2 + (-1)^n}{3 + 2(-1)^n} \right) = 2$

$\therefore$ We can't apply the ~~Ratio~~ Root test whether the series is convergent or divergent.

Example

Consider $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ Try using LART and RT

Important: $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x$